CSE475 — Task 1: Exploratory Data Analysis

NinaPro DB1 EMG + Glove Gesture Recognition Dataset

Group 04

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Abstract

This report presents an exploratory data analysis (EDA) of the NinaPro DB1 surface electromyography (EMG) dataset, combined with its accompanying Cyberglove kinematic recordings, in preparation for gesture classification with classical machine learning baselines and a graph neural network (Task 2) and an architecture ablation study (Task 3). We characterize the dataset's scale, statistical properties, data-quality issues, class balance, feature distributions, inter-feature correlation, and low-dimensional class separability. The central finding of this analysis — a cross-exercise gesture label collision inherent to how NinaPro DB1 numbers its stimuli — is shown to have a direct, measurable effect on downstream classification difficulty and is treated as the dataset's key modeling caveat.

1 Dataset Overview

1.1 A. Summary Table

NinaPro DB1 (https://ninapro.hevs.ch/instructions/DB1.html) is a benchmark surface-EMG dataset for hand and finger gesture recognition, developed for prosthetic-control and human–computer-interaction research. Table 1 summarizes the dataset as loaded and combined for this project.

Table 1: Dataset summary table.

Property	**Value**
Dataset name	NinaPro DB1
Source	https://ninapro.hevs.ch/instructions/DB1.html
Application area	Surface-EMG hand/finger gesture recognition (prosthetic & HCI cont
Subjects	27 (intact)
Exercises (files per subject)	3 (Exercise A/B/C, 81 total `.mat` files)
Sampling rate	100 Hz
Raw signal channels	10 EMG (Otto Bock electrodes) + 22 glove (Cyberglove joint-angle) =
Label column	`restimulus` (refined, post-processed stimulus label)
Raw samples (all subjects, all exercises)	12,553,611 rows
Raw unique label values (incl. rest = 0)	0–23 (24 values)
Host / distribution used	Kaggle mirror: `hoixding/nianpro-db1`
Missing values (raw)	0
Exact duplicate rows (raw)	224,801

The 224,801 raw duplicate rows are not a data-quality defect: NinaPro's rest periods hold a near-constant EMG/glove reading for many consecutive 100 Hz samples while the hand is still, so identical consecutive rows are expected physiological behavior. This does not affect the windowed feature table used for modeling (Section 3), which contains zero duplicate rows.

1.2 Windowed Feature Table

Raw 100 Hz samples are windowed (window size 200 samples, step size 100 samples, i.e. 50% overlap), grouped by (subject, exercise, repetition, gesture) so that no window spans two different gestures or crosses a repetition boundary. Rest (label 0) is excluded from the modeling table. Four time-domain descriptors — RMS, MAV, waveform length (WL), and variance (VAR) — are computed per channel per window, giving $32 \times 4 = 128$ features per window.

Table 2: Windowed feature table composition.

Property	**Value**
Windowed table shape	30,936 rows × 131 columns (128 features + label + subject + exerc
Windows from Exercise A	6,852
Windows from Exercise B	9,435
Windows from Exercise C	14,649
Number of gesture classes (rest excluded)	23

2 B. Descriptive Statistics

Table 3 reports mean, median, standard deviation, min/max, quartiles, skewness, kurtosis, and outlier rate (IQR rule, 1.5×IQR beyond Q1/Q3) for a representative subset of features. The full 128-feature statistics table is exported as `feature_statistics.csv`.

Table 3: Descriptive statistics for representative features.

Feature	Mean	Median	Std	Min	Max	Q1	Q3	Skew	Kurt
emg_0_RMS	0.527	0.349	0.578	0.002	3.802	0.033	0.823	1.486	2.552
emg_0_WL	4.559	3.310	4.443	0.010	28.365	0.420	7.383	1.011	0.475
glove_0_RMS	135.949	135.154	35.836	28.995	255.000	108.403	163.977	0.034	−0.610
glove_5_VAR	241.834	51.529	488.210	0.000	5840.603	5.966	247.012	4.027	21.365

The most heavily skewed features are variance-type descriptors on EMG channels (`emg_4_VAR`: skewness 27.7, kurtosis 872; `emg_5_VAR`: skewness 12.2, kurtosis 199), reflecting the naturally bursty, heavy-tailed nature of muscle activation signals during rapid gesture transitions.

3 C. Data Quality

- **Missing values:** 0 in the windowed feature table.
- **Exact duplicate rows:** 0 in the windowed feature table (224,801 in the raw signal, explained above as expected rest-period behavior).

- **Near-constant features:** none found (no feature has >99% of its values concentrated at a single value).
- **Highly correlated (redundant) feature pairs** ($|r| > 0.95$): 34 pairs found, overwhelmingly RMS-vs-MAV on the same channel (e.g. `glove_0_RMS` vs. `glove_0_MAV`, $r = 1.000$), since both are amplitude estimators of the same underlying signal. This redundancy directly motivated a feature-selection step (SelectKBest, top 100 of 128 features) in the Task 2 baseline pipeline.
- **Outliers:** an average per-feature outlier rate of 5.93% (IQR rule). EMG/glove signals are naturally bursty around movement onsets, so a nontrivial outlier rate reflects real gesture activity rather than sensor error.

Key Finding: Cross-Exercise Label Collision

NinaPro DB1 numbers its gesture stimuli *locally within each exercise file*: rest is always label 0, but the non-rest gesture indices restart from 1 in each of Exercise A, B, and C. Concatenating all three exercises under the raw `restimulus` column — as this project does, to keep the class count in a single, uniform label column — causes classes 1–17 to represent *different physical gestures* depending on which exercise a given window came from. Only classes 18–23 (present in Exercise C alone) are collision-free.

Table 4: Windows per (label, exercise) pair. Any label with nonzero counts in more than one exercise column is a label collision.

Label	Exercise A	Exercise B	Exercise C
1	606	641	553
2	596	559	579
3	712	550	559
4	540	555	575
5	530	461	629
6	585	456	649
7	578	554	588
8	619	546	615
9	540	607	577
10	489	563	602
11	478	624	674
12	579	562	606
13	0	571	576
14	0	448	664
15	0	465	700
16	0	647	627
17	0	626	727
18	0	0	663
19	0	0	676
20	0	0	736
21	0	0	632
22	0	0	644
23	0	0	798

Modeling implication: windows sharing a collided label are not the same gesture, which adds label noise for classes 1–12 (present in both Exercise A and Exercise B). This is the most likely explanation for the lower, noisier per-class F1 scores observed for classes 1–12 relative to the collision-free classes 18–23 in the Task 2 results, and is treated throughout this project as an explicit scoping/label-design limitation rather than a modeling failure.

4 D. Class Balance

Table 5: Windowed sample count per gesture class (rest excluded).

Class	1	2	3	4	5	6	7
Count	1800	1734	1821	1670	1620	1690	1720
Class	8	9	10	11	12	13	14
Count	1780	1724	1654	1776	1747	1147	1112
Class	15	16	17	18	19	20	21
Count	1165	1274	1353	663	676	736	632
Class	22	23					
Count	644	798					

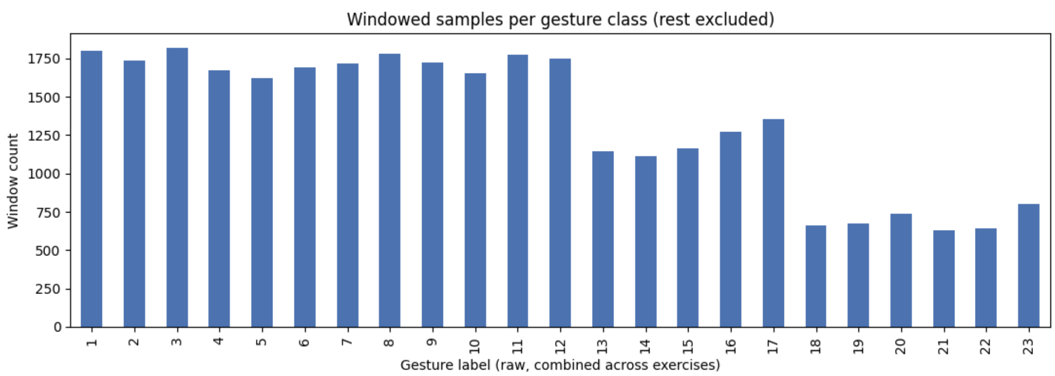

Figure 1: Windowed samples per gesture class (bar chart).

The imbalance ratio between the largest and smallest class is 2.88× (1800 for class 1 vs. 632 for class 21). This moderate imbalance — compounded by the label collision inflating classes 1–12's effective sample counts with mixed-source data — motivates reporting Macro-F1 alongside accuracy in Task 2: plain accuracy would over-reward performance on the frequent, collision-affected classes and could mask poor performance on the rarer, collision-free classes.

5 E. Feature Distributions

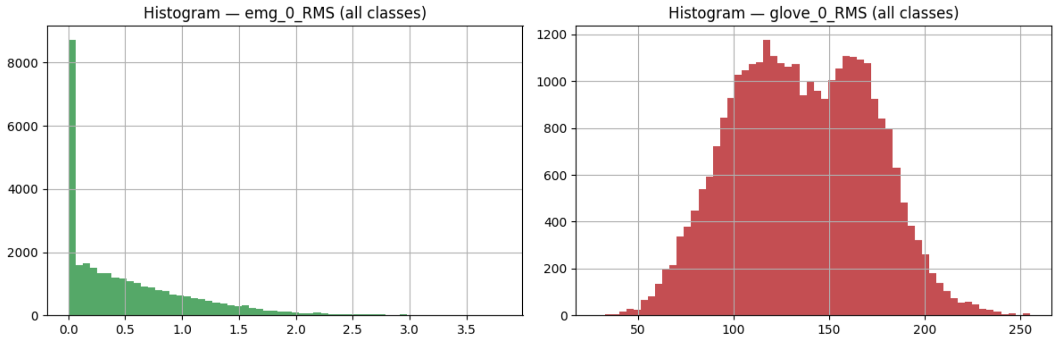

Figure 2: Histograms of a representative EMG feature (emg_0_RMS) and glove feature (glove_0_RMS) across all classes.

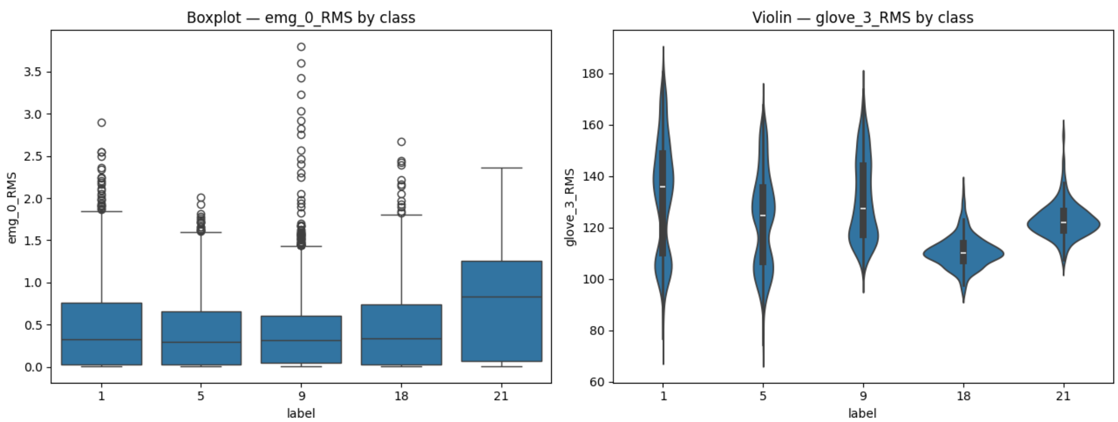

Figure 3: Boxplot of emg_0_RMS and violin plot of glove_3_RMS by class, for a representative subset of classes (a mix of collision-affected classes 1, 5, 9 and collision-free classes 18, 21).

Reading: glove features show visibly different medians and spreads across classes, since the glove is a near-direct pose readout. Single EMG features, by contrast, overlap heavily across classes — no individual EMG channel/feature cleanly separates gestures on its own, which motivates the use of multi-channel graph-based or ensemble ML models rather than simple per-feature thresholding.

6 F. Correlation Analysis

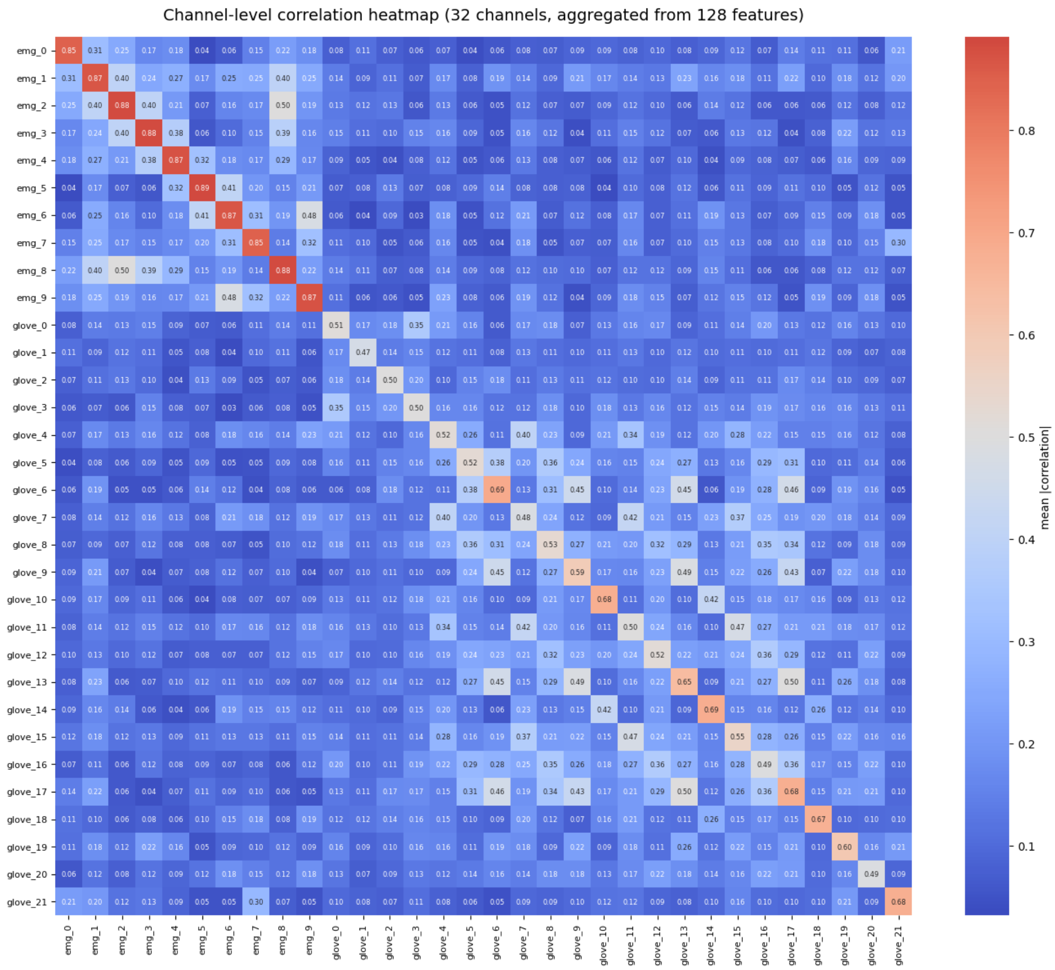

Figure 4: Channel-level correlation heatmap (32 channels, each feature aggregated by mean absolute correlation across its four descriptors) — a readable summary of the full 128-feature correlation structure.

34 feature pairs exceed $|r| > 0.95$, dominated by RMS-vs-MAV pairs on the same channel (both are amplitude estimators of the same signal, so near-perfect correlation is expected). The full, unannotated 128×128 feature-level heatmap is retained as a supplementary reference figure (`correlation_heatmap_full.p`

7 G. 2D/3D Projections: PCA, t-SNE, UMAP

Projections were computed on a class-stratified subsample (5,980 windows) of the standardized 128-dimensional feature space.

Table 6: PCA explained variance (first 3 components).

Component	PC1	PC2	PC3	Cumulative
Explained variance ratio	0.1755	0.1026	0.0688	0.3468

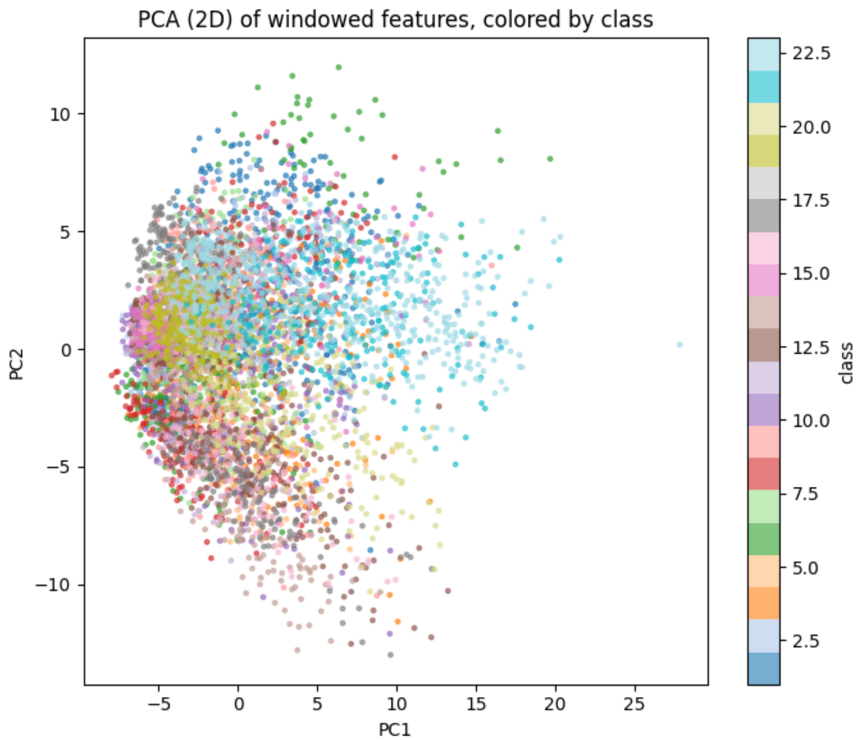

Figure 5: PCA (2D) projection, colored by class.

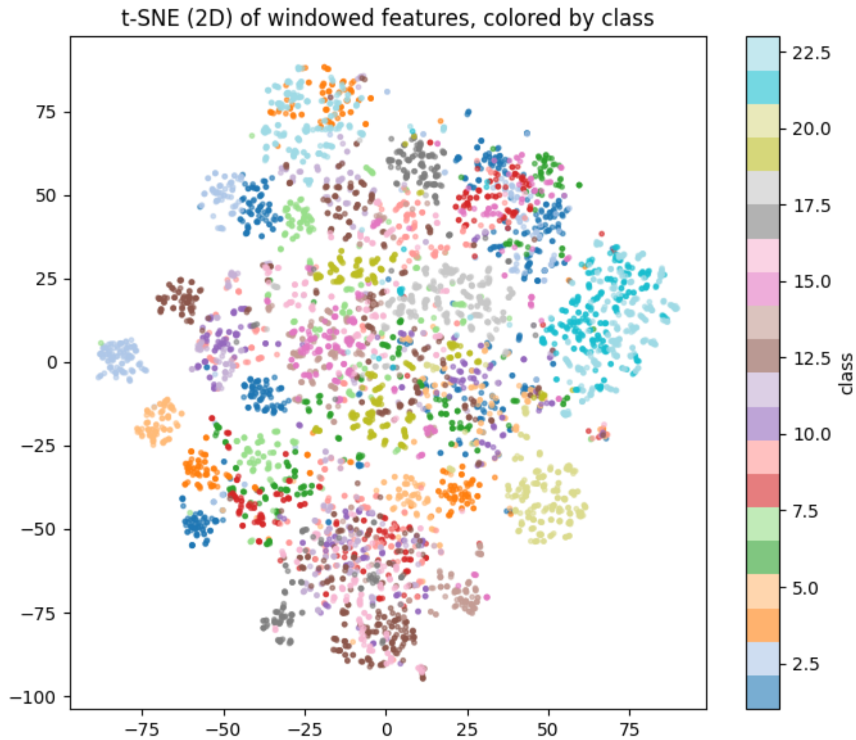

Figure 6: t-SNE (2D) projection, colored by class.

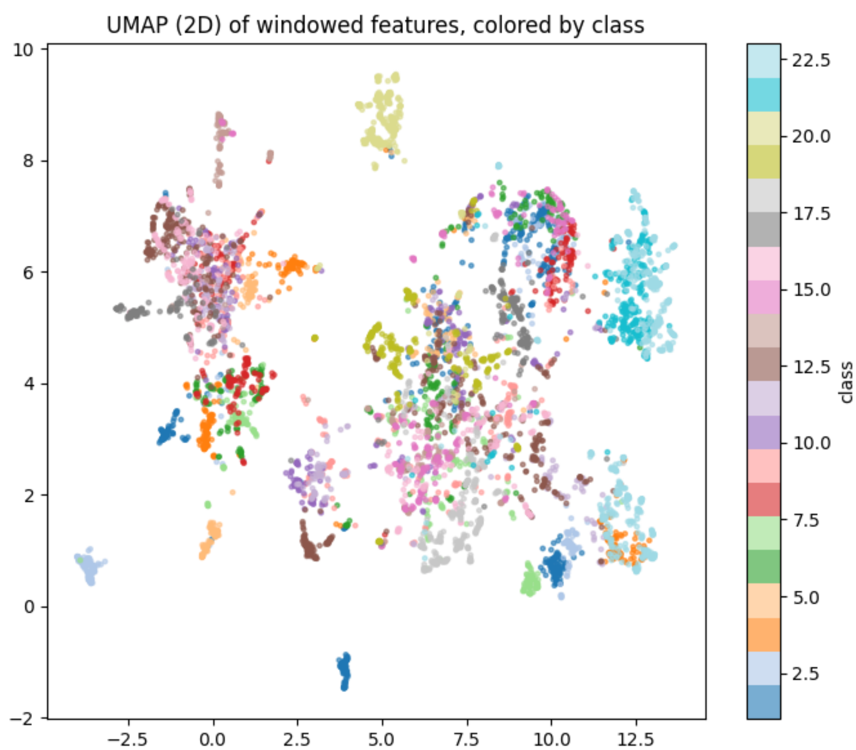

Figure 7: UMAP (2D) projection, colored by class.

Reading: PCA's first three components capture only 34.7% of total variance and show broad, overlapping clusters rather than clean separation — consistent with (a) 23 classes being difficult to separate linearly, and (b) the label collision blurring boundaries between Exercise-A/B classes. The nonlinear t-SNE and UMAP projections show tighter, more numerous local clusters; in both, collision-free classes (18–23) tend to form more distinct islands than the collision-affected classes (1–17), directly corroborating the Section C finding.

8 H. Interactive Plots

Interactive (Plotly) versions of the class-balance bar chart and PCA scatter plot were produced to allow hover-based inspection of exact class counts and individual point identities, and per-class isolation via the legend. These are included in the accompanying Jupyter notebook (`Group04_NinaProDB1_task1_eda.i`
as live, interactive figures rather than static images, since interactivity does not render in a compiled PDF.

9 Summary of EDA Findings

1. **Scale:** 27 subjects × 3 exercises (81 raw files, 12,553,611 raw 100 Hz samples) windowed into 30,936 labeled samples across 23 gesture classes (rest excluded).

2. **No missing values** and **no exact duplicates** in the windowed feature table; raw-signal duplicates (224,801 rows) are explained by expected rest-period signal stability.

3. **Moderate class imbalance** (2.88×) motivates Macro-F1 as the primary Task 2 metric.

4. **34 redundant feature pairs** ($|r| > 0.95$) motivate feature selection in the Task 2 baseline pipeline.

5. **Cross-exercise label collision** (classes 1–17 mix two distinct physical gestures each; classes 18–23 are collision-free) is the dataset's most consequential structural property for this project, and is the leading explanation for the per-class performance gap observed between collision-affected and collision-free classes throughout Task 2 and Task 3.

6. **Glove features separate classes far more cleanly than EMG features** in both univariate distribution plots and the low-dimensional projections — expected, since glove records joint angle nearly directly, while EMG is an indirect muscle-activity proxy. This motivates the explicit caveat, carried into the Task 2 report, that EMG+glove results are an easier formulation than a real-world EMG-only prosthetic-control setting.